KARAKTERISTIEKEN

VOOR SILLYCON SYMPOSIA

- account aanmaken
- "jokes" en "puns" maken
- berichten tot 281 karakters posten
- links posten
- "Haha" of "Giggle" reageren
- een eigen icoontje
- sprekers kunnen een forum opzetten over hun eigen topic
- ondersteuning voor internationaal gebruik
- kleine support staff
- pieken in verkeer (tijdens conferenties)
- scalability (veel users)
- elasticity (pieken)
- integrity (accounts bepalen ownership)
- internationalization (zie context)
- customizability (aanmaak forums, icoontjes)
- fault tolerance (kleine support staff)

DOCUMENTEREN

Karakteristiek	Expliciet?	Top 3?
Scalability	Ja	**Ja**
...		
Fault tolerance	Nee	**Ja**

ARCHITECTURALE BESLISSINGEN

TWEEDE WET

Why is more important than **how**.

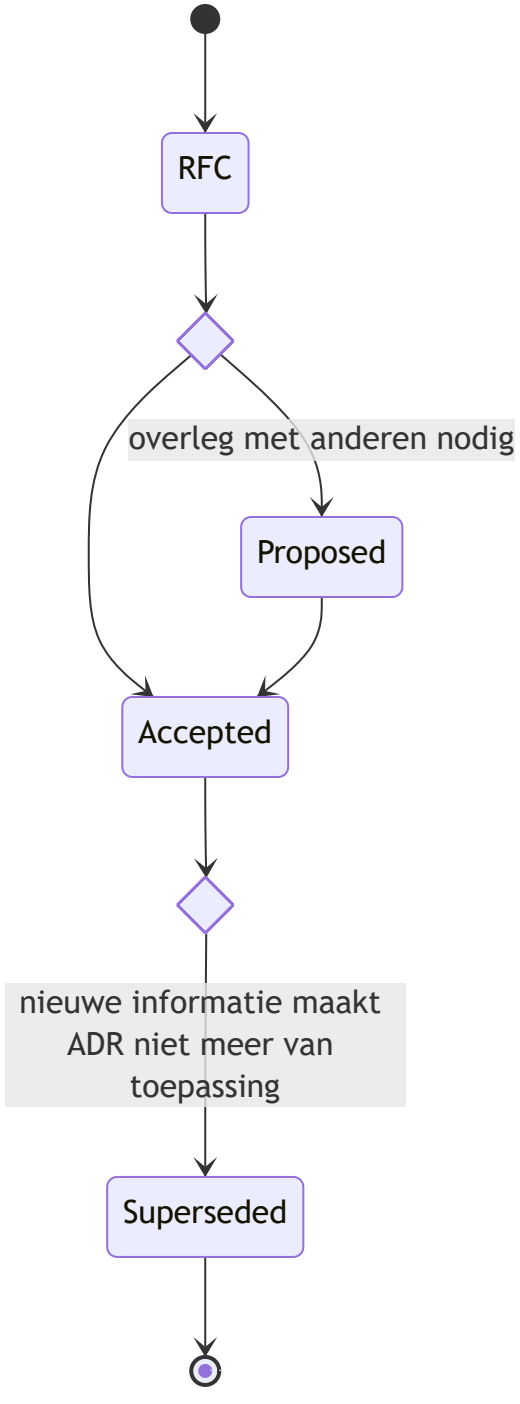
ADR'S

- één per architecturale beslissing
- gestructureerd
- geen "user-facing" docs
- append-only
- best plain text
- typisch per project
- neutrale toon

MOGELIJKE TEMPLATE

- title
- status
- context
- decision
- consequences
- governance
- notes

TOESTANDSDIAGRAM


RFC
overleg met anderen nodig
Proposed
Accepted
nieuwe informatie maakt ADR niet meer van toepassing
Superseded

- Title: 012: Gebruik van queues voor asynchrone messaging vanaf tradingdienst
- Status: Accepted
- Context: De trading service moet andere diensten (voorlopig: analytics en notifications) op de hoogte brengen van nieuwe producten en van elke verkoop. Dit kan via web services (REST e.d.) of via asynchrone messaging (queues of topics).
- Decision: **We zullen queues gebruiken.** Queues maken het systeem veelzijdig, aangezien elke queue andere soorten berichten kan afleveren. Ze maken het systeem ook veiliger aangezien de trading service steeds weet wie de queues uitleest.

extra tools voor ADR

LOGISCHE COMPONENTEN

STAP 1: INITIËLE KERNCOMPONENTEN

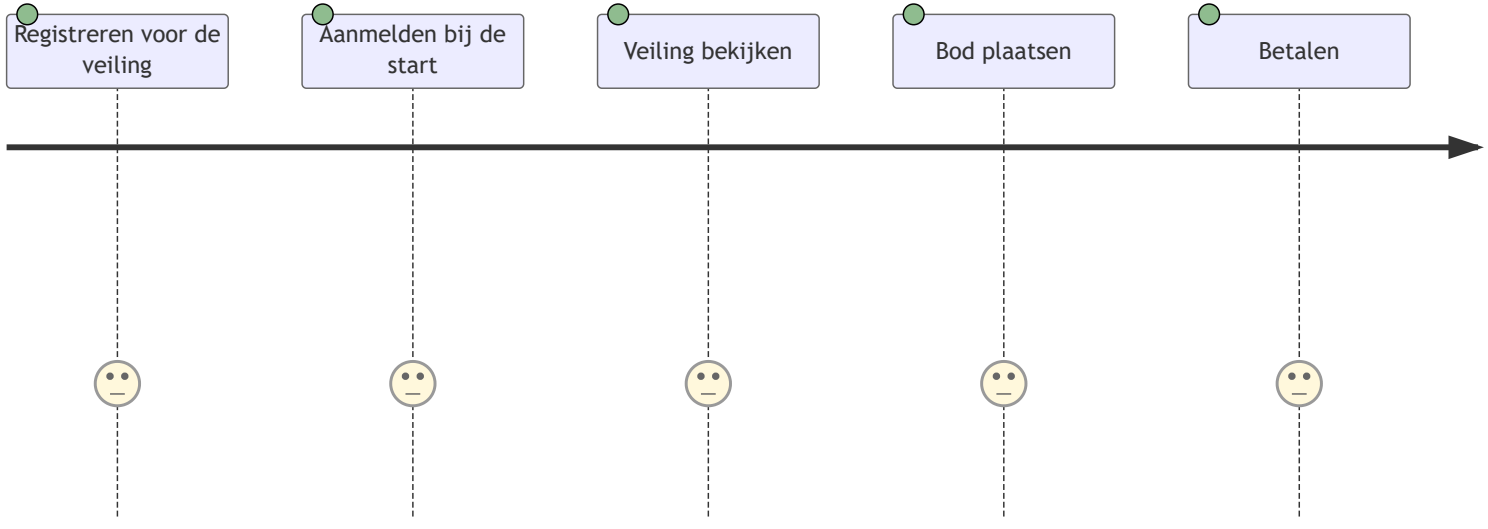
- 2 technieken
  - workflow
  - actor/action
- mogen beide toepassen

WORKFLOW APPROACH

- perspectief van een gebruiker
- conceptuele stappen van één "user journey"
- verschillende stappen ⇒ verschillende componenten
  - niet nodig voor sterk verwante stappen
  - omgekeerd: één stap kan meerdere aspecten bevatten en dus beter over meerdere componenten verdeeld worden

"Meedoen en winnen"

Bieder



NAMEN

- Auction Registration
- Live Auction Session
- Video Streamer
- Bid Capture
- Automatic Payment

ACTOR/ACTION APPROACH

- handig bij meerdere soorten gebruikers
- sommen **belangrijkste** handelingen op
- verbinden deze met componenten
- verbinden ook componenten onderling
- "system" actor voor automatische acties

Remember Kate? She wants to go on a trip, so she'll be a bidder.

This is Sam. He's an auctioneer.

Look for an auction

View the live video stream

Place a bid

Receive a bid from an online bidder

Enter a live in-person bid into the system

Start the auction

Mark an item as sold

Auction Search

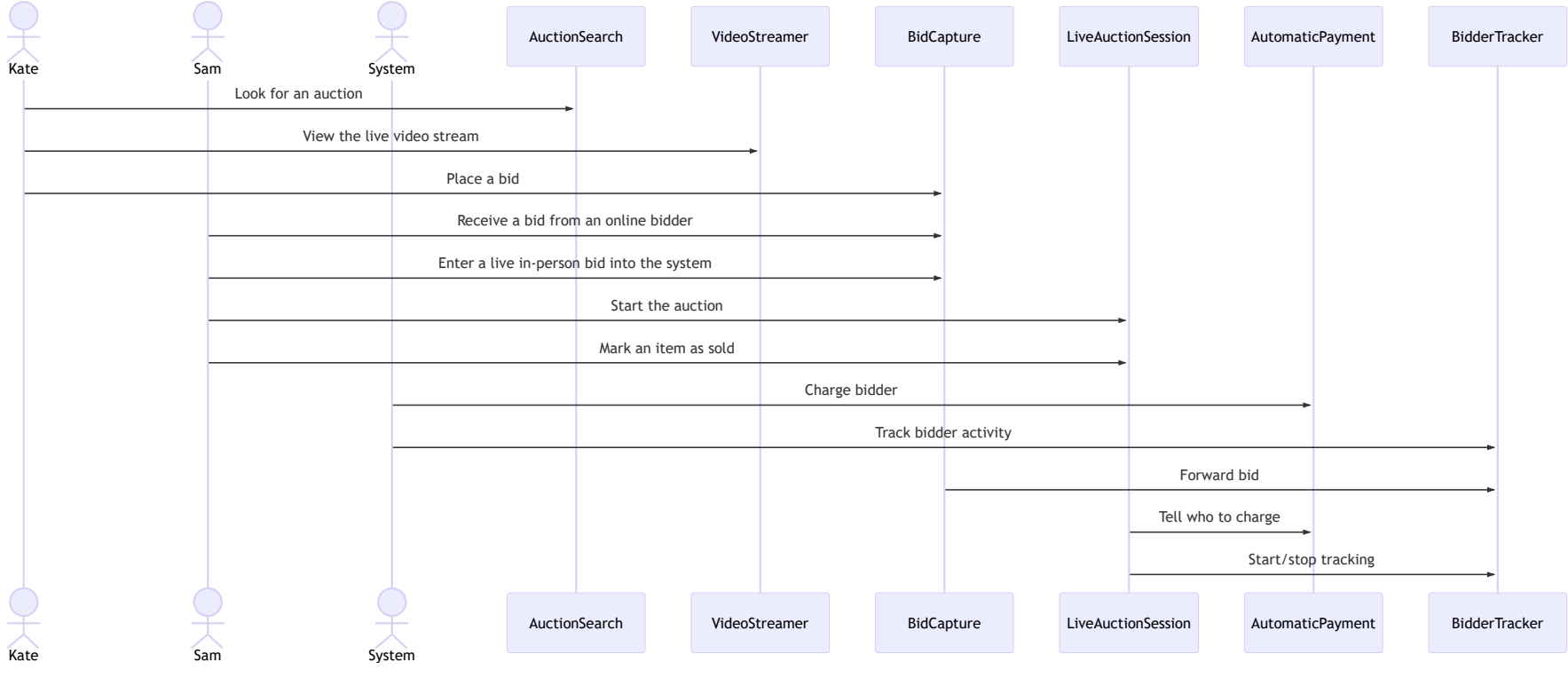
Video Streamer

Bid Capture

Live Auction Session

The Bid Capture component handles multiple actions and sends bids to the Bidder Tracker component.

The Live Auction Session component sends bidder information to other components.


Kate
Sam
System
AuctionSearch
VideoStreamer
BidCapture
LiveAuctionSession
AutomaticPayment
BidderTracker
Look for an auction
View the live video stream
Place a bid
Receive a bid from an online bidder
Enter a live in-person bid into the system
Start the auction
Mark an item as sold
Charge bidder
Track bidder activity
Forward bid
Tell who to charge
Start/stop tracking
Kate
Sam
System
AuctionSearch
VideoStreamer
BidCapture
LiveAuctionSession
AutomaticPayment
BidderTracker

ENTITY TRAP

- component met twee verantwoordelijkheden
- extra risico bij vage naamgeving
  - "supervisor"
  - "manager"
  - "control center"
  - ...

COMBINEREN

- identificeer actors en primaire acties zoals in actor/action
- doorloop dan acties zoals in workflow
- kunnen zo meer in detail gaan indien zinvol

STAP 2: REQUIREMENTS TOEWIJZEN AAN COMPONENTEN

Display the current trip up for bid during the auction

Since the Live Auction Session component is responsible for controlling the auction currently taking place, it makes sense to assign this requirement to it.

Mark a trip as sold and move to the next trip

This requirement also has to do with controlling the flow of the auction session, so let's assign it to the Live Auction Session component.

Send each online bid to the live auctioneer as it's placed, so they can call out the bid

Since the Bid Capture component receives the bid from the bidder for a trip, it makes sense to have this component also forward that bid to the live auctioneer.

Record the live video stream for later playback in case a bid is contested

The Video Streamer component already has the video feed, so it makes sense to have it record the video feed as well.

Allow bidders to view details about the current trip up for bid

Wait—it doesn't make sense for any of the other components we identified earlier to take on this responsibility. Let's create

Represents the live auction currently taking place

Responsible for receiving and storing bids from the bidders.

Responsible for showing the video feed from the auction

This is a new component we added to satisfy the last requirement

Creating new components is known as evolving the architecture.

STAP 3: ROL EN VERANTWOORDELIJKHEDEN ANALYSEREN

- per component vraag stellen: "welke taken?"
- interne samenhang van een component = "cohesie"
  - hoe duidelijker het takenpakket, hoe hoger de cohesie
- moet opgevolgd worden naarmate systeem groeit
  - kan introductie nieuwe componenten noodzakelijk maken
    - kan bijhouden in een ADR!
- onthoud: **mik op sterke cohesie**

STAP 4: ARCHITECTURALE KARAKTERISTIEKEN ANALYSEREN

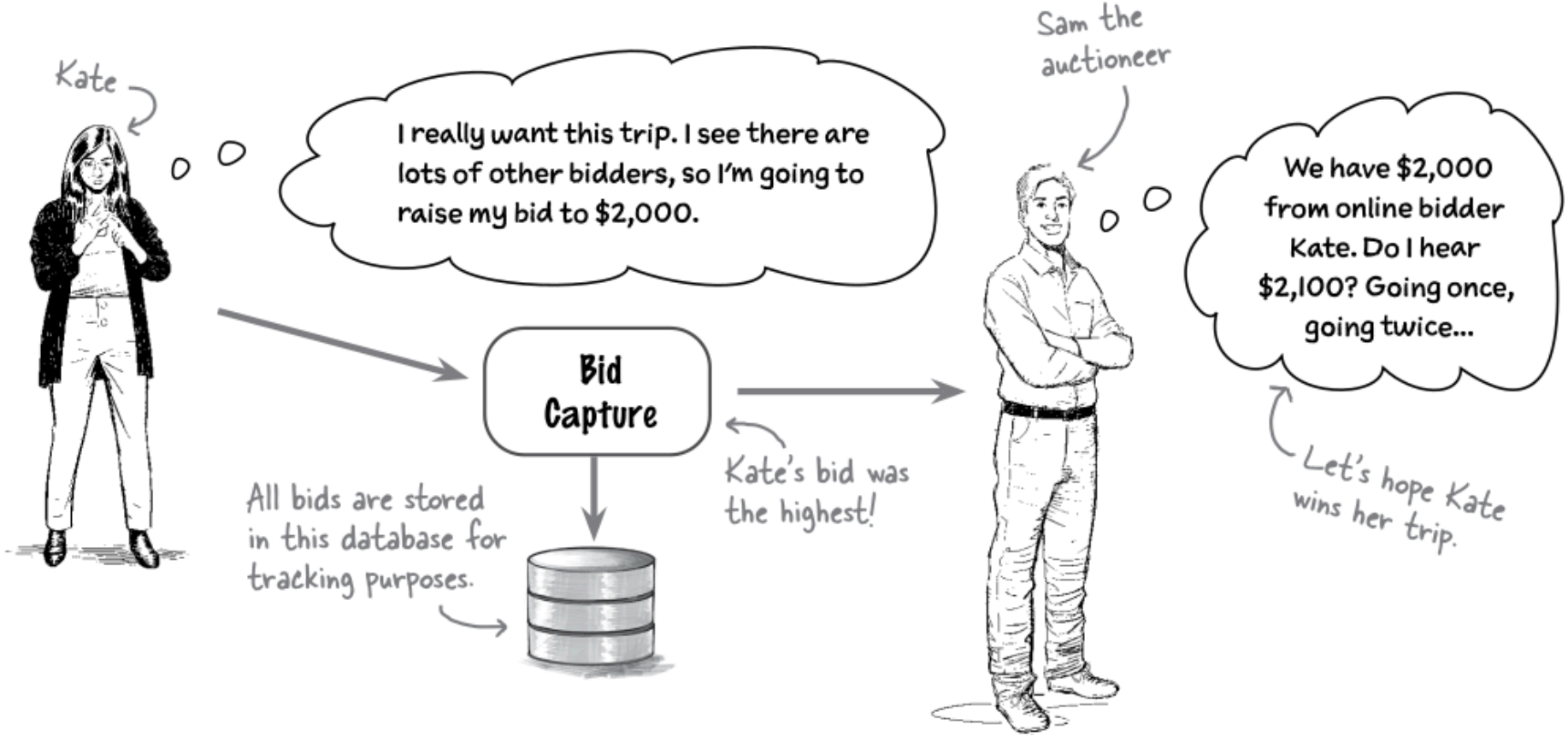
- kijken hier naar de "driving characteristics"
- bekijken of de takenverdeling deze ondersteunt
- hier wel idee van fysieke implementatie nodig
  - moet convergeren wat betreft eerste beslissingen, componenten en stijl

VOORBEELD (1)

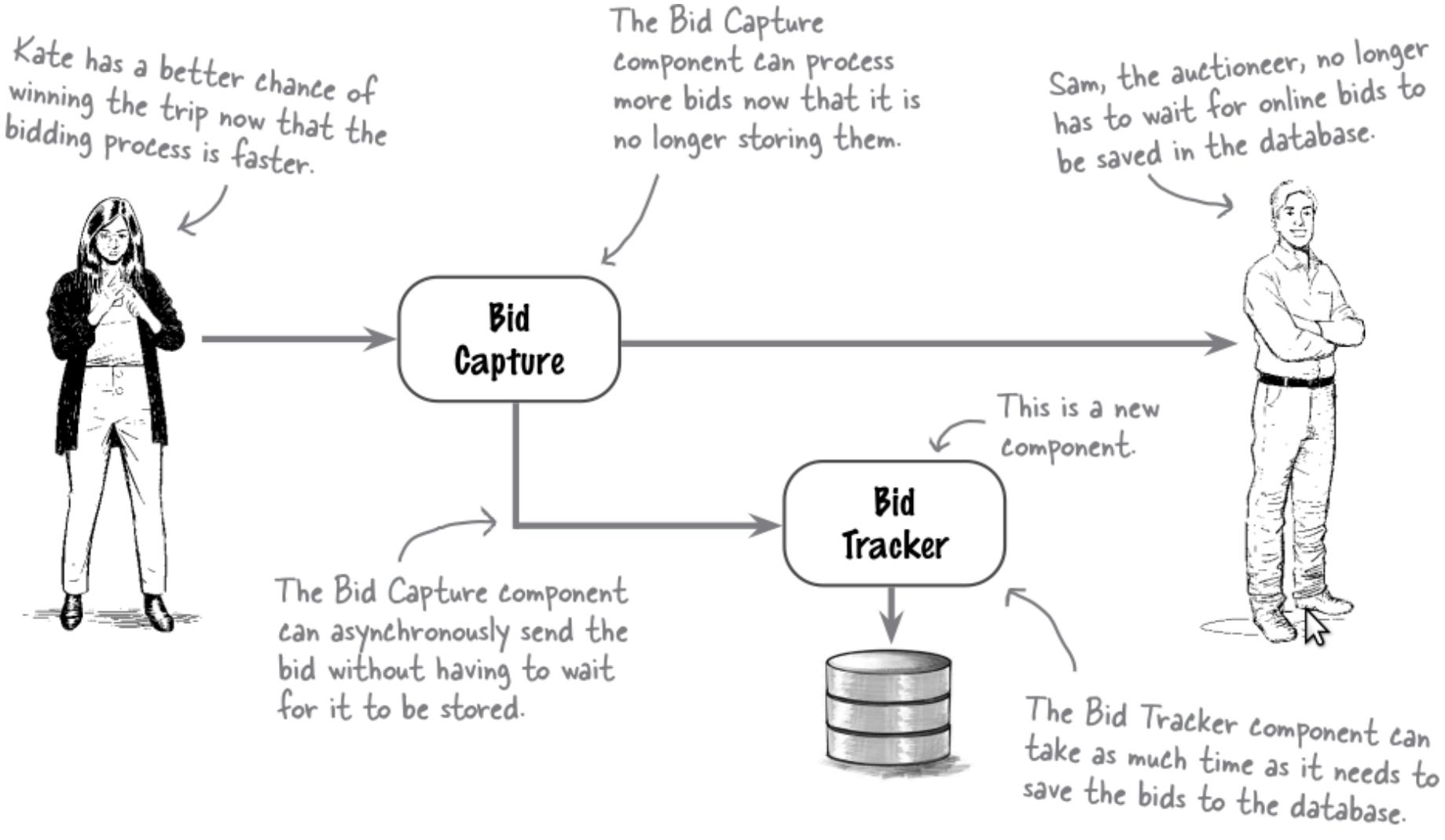
driving characteristics:

- scalability
- availability
- performance

VOORBEELD (2)

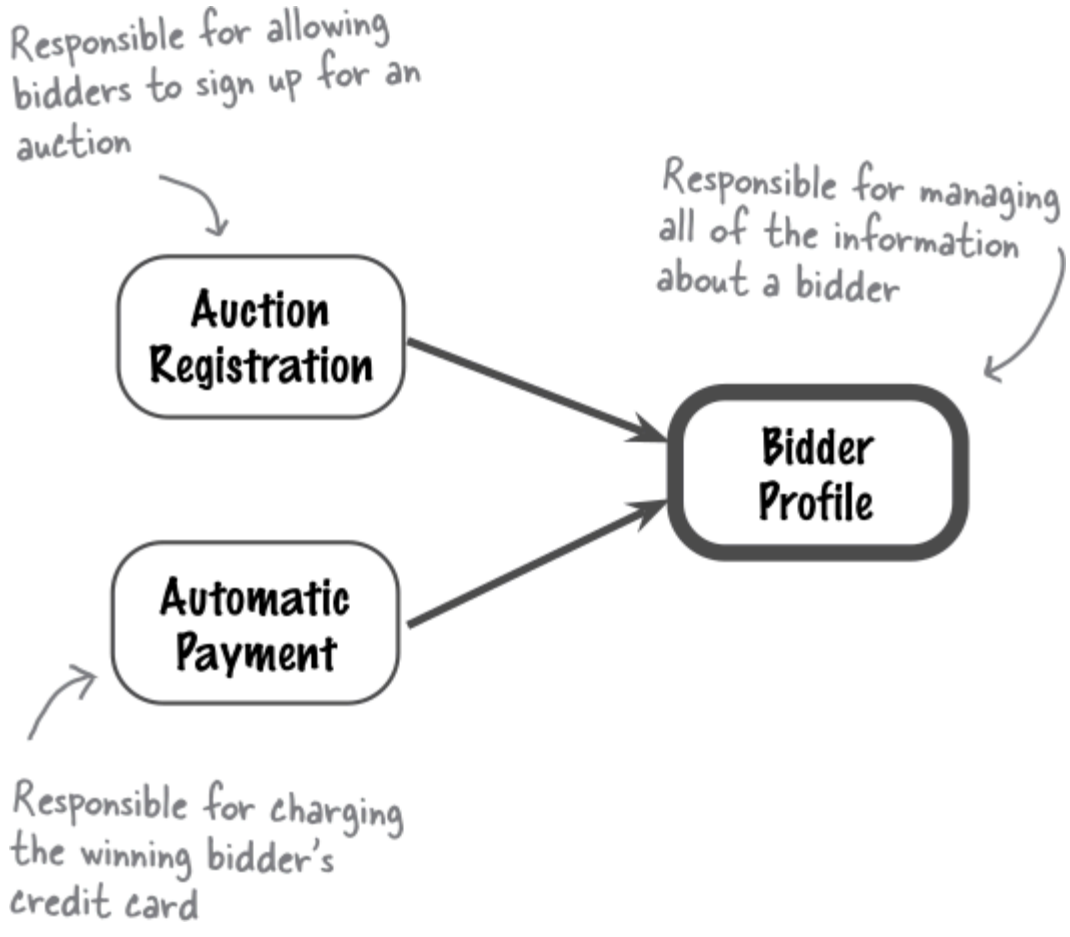

Kate
I really want this trip. I see there are lots of other bidders, so I'm going to raise my bid to $2,000.
Bid Capture
All bids are stored in this database for tracking purposes.
Kate's bid was the highest!
Sam the auctioneer
We have $2,000 from online bidder Kate. Do I hear $2,100? Going once, going twice...
Let's hope Kate wins her trip.

VOORBEELD (3)

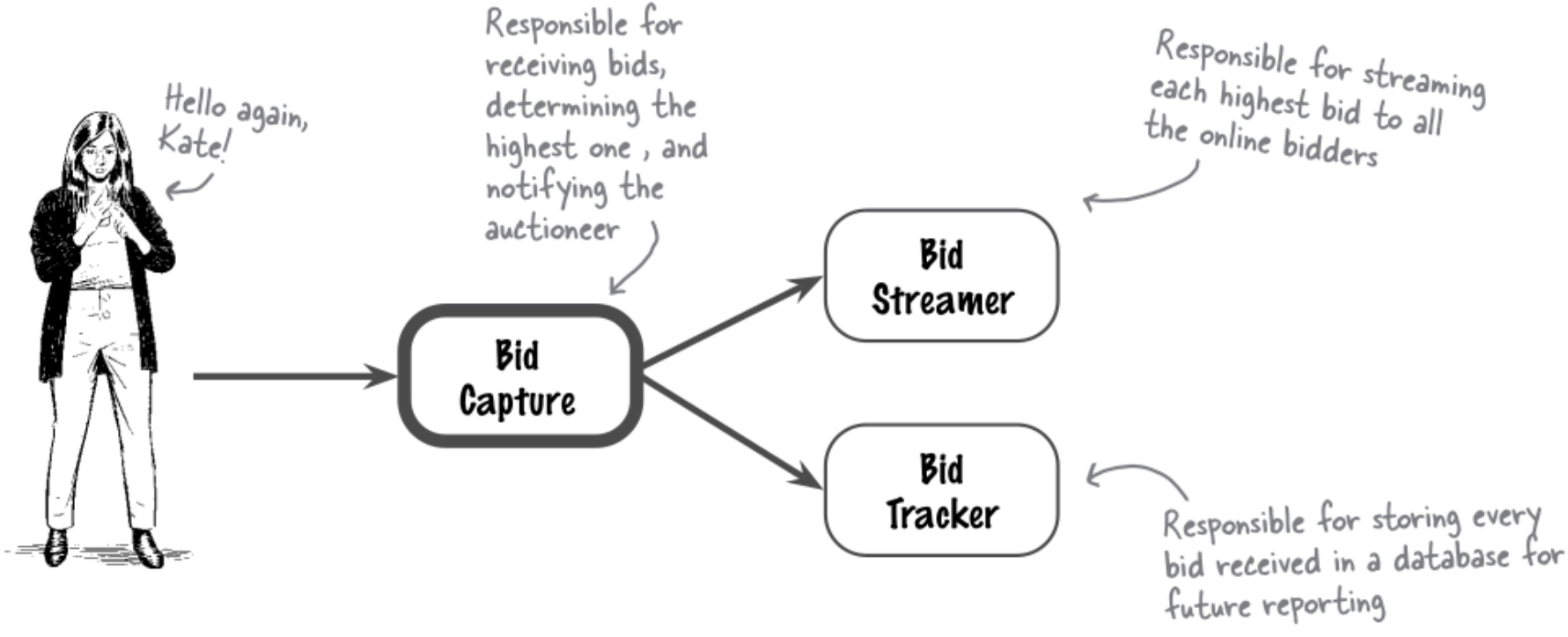

Kate has a better chance of winning the trip now that the bidding process is faster.
The Bid Capture component can process more bids now that it is no longer storing them.
Sam, the auctioneer, no longer has to wait for online bids to be saved in the database.
Bid Capture
This is a new component.
Bid Tracker
The Bid Capture component can asynchronously send the bid without having to wait for it to be stored.
The Bid Tracker component can take as much time as it needs to save the bids to the database.

INTERACTIE TUSSEN COMPONENTEN

AFFERENTE KOPPELING


Responsible for allowing bidders to sign up for an auction
Responsible for managing all of the information about a bidder
Auction Registration
Bidder Profile
Automatic Payment
Responsible for charging the winning bidder's credit card

EFFERENTE KOPPELING

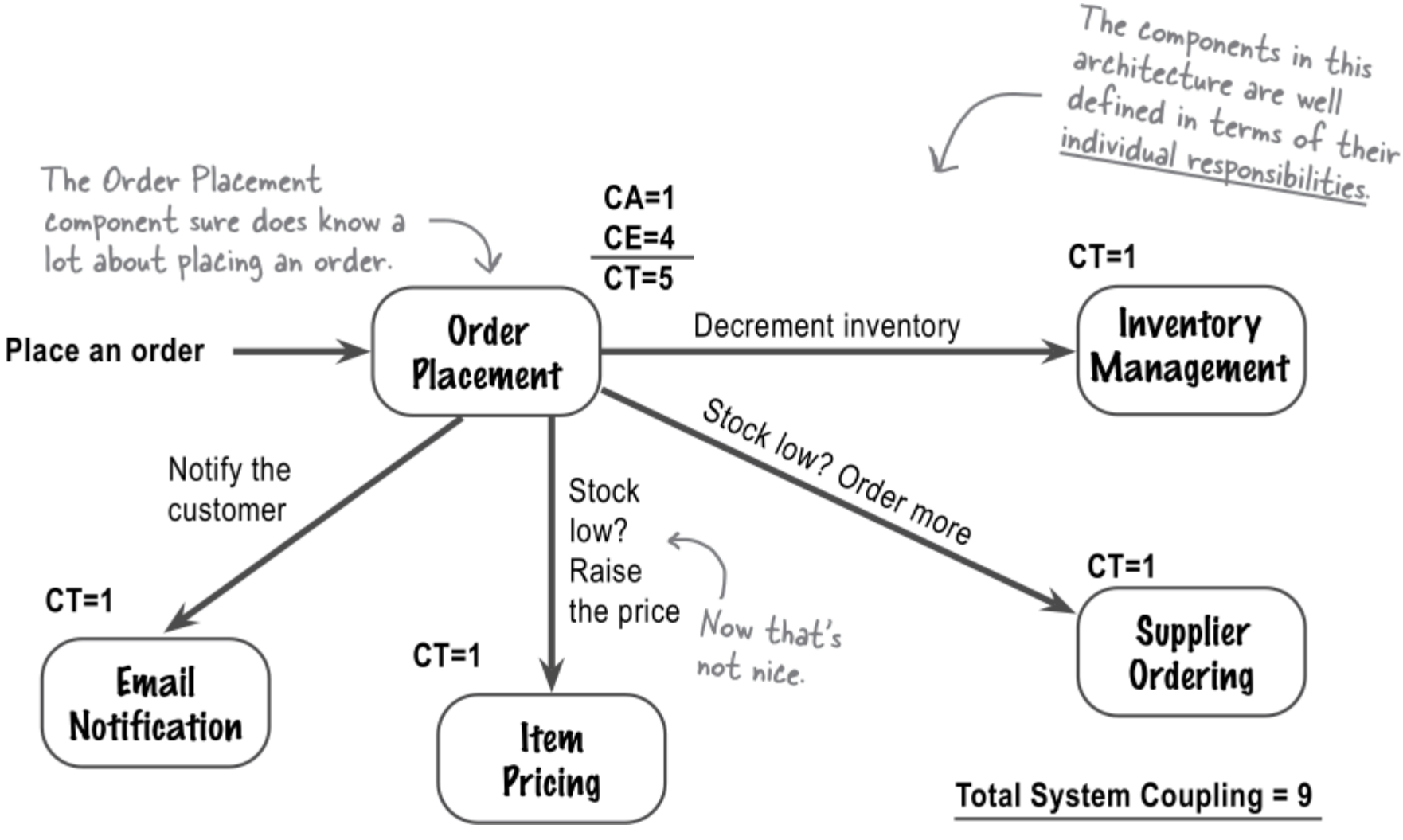

Hello again, Kate!
Responsible for receiving bids, determining the highest one , and notifying the auctioneer
Bid Capture
Bid Streamer
Bid Tracker
Responsible for streaming each highest bid to all the online bidders
Responsible for storing every bid received in a database for future reporting

KOPPELING METEN

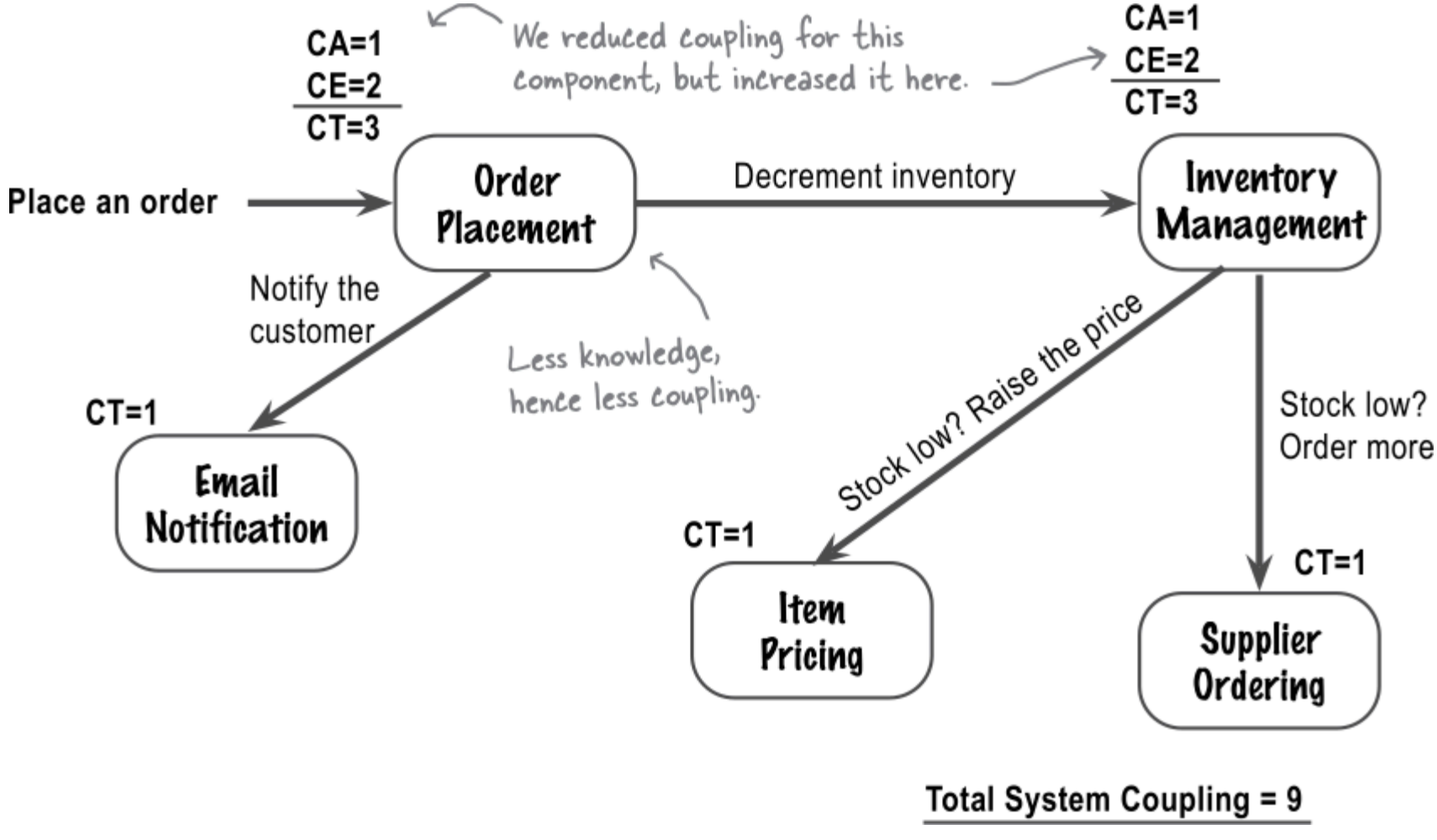
- per component afferente, efferente en totale (som) koppeling
- ook totale koppeling van het systeem (som van sommen)

WET VAN DEMETER

TOEGEPAST (1)


The components in this architecture are well defined in terms of their individual responsibilities.
The Order Placement component sure does know a lot about placing an order.
CA=1
CE=4
CT=5
CT=1
Place an order
Order Placement
Decrement inventory
Inventory Management
Stock low? Order more
Notify the customer
Stock low? Raise the price
Now that's not nice.
CT=1
CT=1
CT=1
Email Notification
Item Pricing
Supplier Ordering
Total System Coupling = 9

TOEGEPAST (2)


CA=1
CE=2
CT=3
We reduced coupling for this component, but increased it here.
CA=1
CE=2
CT=3
Place an order
Order Placement
Decrement inventory
Inventory Management
Notify the customer
Less knowledge, hence less coupling.
Stock low? Raise the price
Stock low? Order more
CT=1
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CT=1
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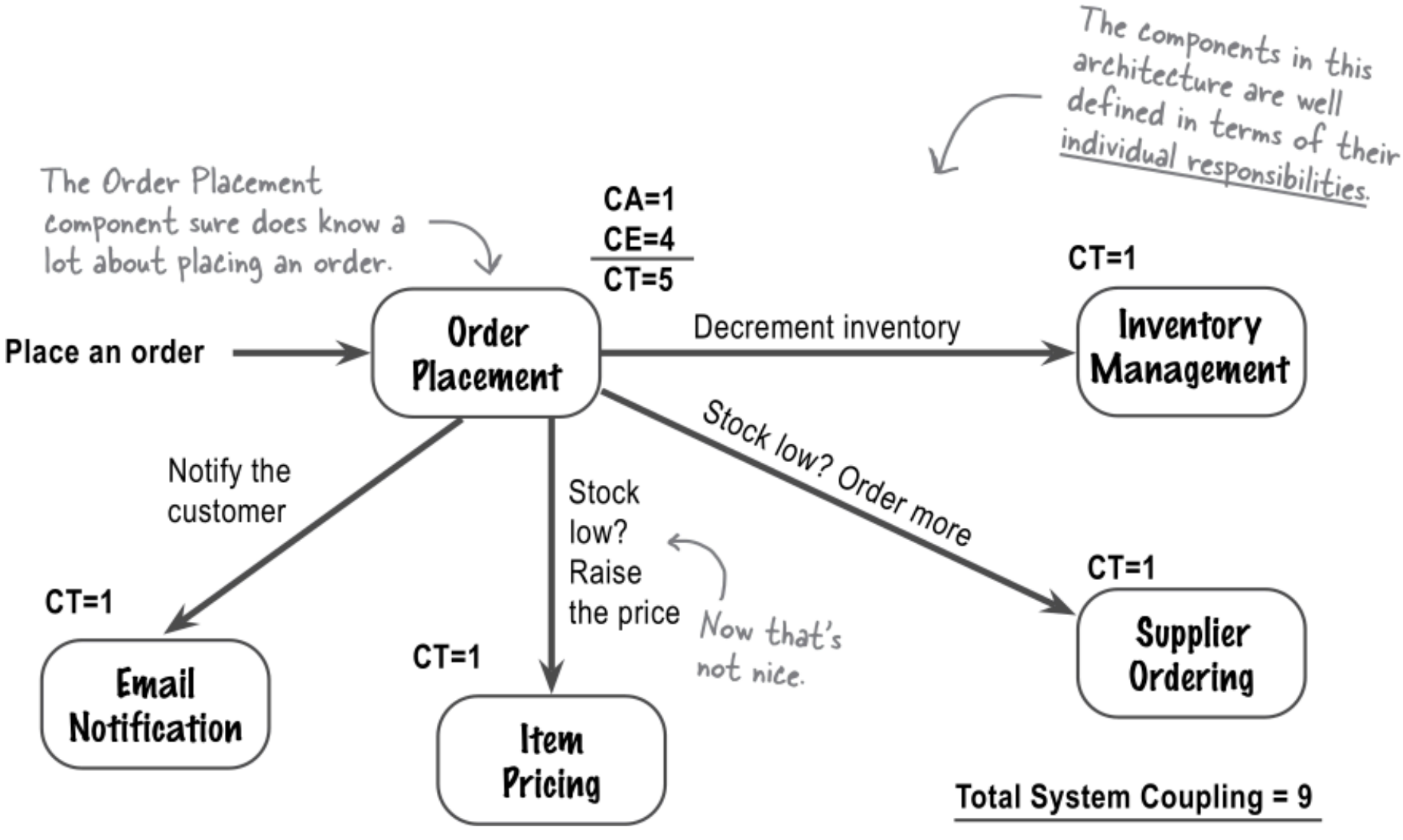
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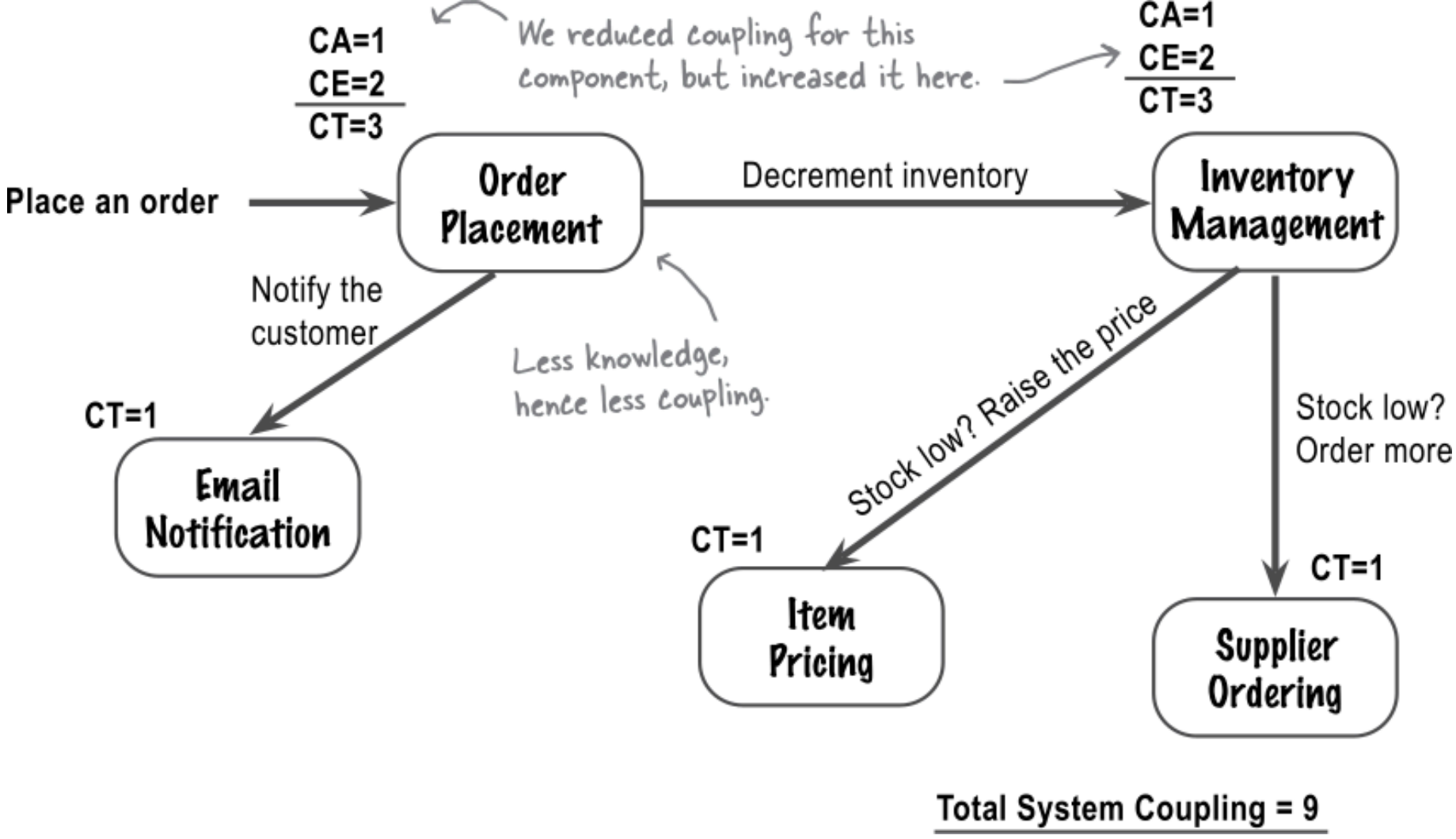
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